

Medicare Billing Risk & Anomaly Detection Dashboard

Project Proposal for Power BI + Machine Learning

One page Summary

One sentence concept: Build a Power BI dashboard using official CMS Medicare claims-derived billing data and an unsupervised machine-learning model that assigns providers a peer-relative anomaly risk score, helping analysts prioritize unusual billing patterns for human review.

Core Question: how can machine learning and Power BI be used to identify unusual Medicare billing patterns and prioritize providers for further review.

Data Source: CMS Medicare Physician & Other Practitioners data based on actual Medicare administrative claims. Official government data.

Main search: [link](#)

Dataset that will be used: [link](#)

Decision	Proposed Choice	Why
Domain	Medicare Part B billing analysis	Public, high-value insurance/claims context with and official federal data source
Dataset	CMS Medicare Physician & Other Practitioners	Traceable to data.cms.gov
ML type	Unsupervised anomaly detection	Public dataset does not contain a verified fraud label, so supervised fraud classification would not be working
Primary Model	Isolation Forest	Designed to rank unusual observations without requiring fraud labels
ML output	Billing anomaly Risk Score, 0-100	A triage signal, not declaration of fraud
BI purpose	Investigation prioritization and explainability	Turns model output into a decision workflow rather than just a static visualization

Important academic boundary

The project will never label a provider as fraudulent based only on this dataset. It will identify unusual billing behavior that may warrant further review. The distinction between anomaly and fraud will be stated throughout the dashboard and report.

1. Executive Summary

This project is a decision-support project that combines official Medicare billing data, machine learning, and Power BI. The project will analyze billing patterns in the Centers for Medicare & Medicaid Services (CMS) Medicare Physician & Other Practitioners - by Provider and Service public dataset. CMS describes the file as information on Original Medicare fee-for-service Part B services and procedures, organized by provider, HCPCS service code, and place of service. The latest available year is 2024 and the dataset is updated annually.

The machine-learning component will not attempt to prove fraud. Because the public file does not contain a validated fraud outcome, the project will use unsupervised anomaly detection to identify provider-service billing patterns that differ materially from appropriate peer groups. The model will generate a 0-100 Billing Anomaly Risk Score. Power BI will convert that score into a review queue, peer comparisons, provider drill-through pages, and model transparency views.

The goal is to demonstrate an end-to-end analytics workflow: source validation, data preparation, domain-aware feature engineering, machine learning, explainability, BI design, and responsible interpretation. The final product is intended as a class project and portfolio piece rather than a production fraud-detection system.

2. Business Problem and Project Objective

Insurance and public health programs process large volumes of billing activity. Analysts cannot manually examine every provider-service pattern with equal attention. A useful analytics system should therefore help prioritize where limited review capacity should be focused while preserving human judgment.

2.1. Business Problem and Project Objective

How can machine learning and Power BI be combined to identify unusually high-risk Medicare billing patterns, compare them with relevant peers, and prioritize provider-service records for further analyst review?

2.2. Business Problem and Project Objective

The project will develop a reproducible pipeline that:

- Uses one official, traceable CMS public dataset as the sole analytical data source.
- Transforms published billing and utilization variables into peer-aware behavioral features.
- Applies an unsupervised ML model to generate an anomaly-based risk ranking.

- Explains the main factors contributing to a high-risk score.
- Integrates the results into an interactive Power BI dashboard for investigation prioritization.
- Communicates model limitations so that anomaly is not misrepresented as confirmed fraud.

3. Research questions

1. Which Medicare provider-service billing patterns are most unusual relative to comparable peers?
2. Can an unsupervised machine-learning model convert multiple billing indicators into a useful anomaly risk ranking?
3. Which features most frequently drive high-risk rankings, such as service intensity, submitted-charge ratios, or unusual procedure concentration?
4. How can Power BI make the model actionable by allowing an analyst to prioritize a limited share of providers for review?
5. How stable are the highest-risk rankings when model assumptions, contamination levels, or feature sets change?

4. Machine-Learning/AI Methodology

4.1. Why Unsupervised Learning

The CMS public file does not provide a verified target variable such as Fraud = Yes/No. Creating a fake target from arbitrary rules would undermine the project. Therefore, the main ML task will be anomaly detection: identifying observations that are unusual relative to comparable billing patterns.

Model interpretation

High anomaly score = unusual billing pattern relative to peers. It does NOT equal proven fraud, abuse, waste, poor care, or criminal conduct.

4.2. Primary Model: Isolation Forest

Isolation Forest will be the primary model because it is designed for anomaly detection without labeled outcomes. It isolates observations using randomly selected features and split values; observations that are isolated unusually quickly receive stronger anomaly scores. This makes it appropriate for ranking rare or unusual provider-service patterns. The model will produce a continuous anomaly score. For Power BI, that score will be transformed into an intuitive 0-100 Billing Anomaly Risk Score, where higher values indicate more unusual peer-relative behavior.

4.3. Baseline/Comparison Models

To show that the result is not dependent on a single algorithm, the project will compare the Isolation Forest ranking with at least one transparent baseline:

- Peer-relative robust z-scores / percentile flags for key features; and/or
- Local Outlier Factor (LOF) as a second unsupervised anomaly method if dataset size and runtime allow.

The final model will be selected based on ranking stability, interpretability, computational feasibility, and usefulness in the Power BI workflow rather than novelty alone.

5. Model Evaluation

Validation Test	Question Answered		Main Metric
Ranking Stability	Do the same observations remain near the top when random seeds or reasonable model settings change?	Re-run model many times with different random seeds	Spearman correlation + Top K overlap
Sensitivity Analysis	How much does the top-risk set change when contamination assumptions or feature sets change?	Change feature sets, peer definitions, preprocessing, model settings	Score/rank changes + Top K overlap
Model Agreement	Do Isolation Forest and transparent percentile/z-score methods identify overlapping high-risk patterns?	Compare isolation forest against peer-percentile/z-score baseline	Rank correlation + overlap
Top-K profiling	What billing characteristics distinguish the top 1%, 5%, and 10% anomaly-risk groups from the overall dataset?	Compare top risk groups with the rest of data	Median differences, percentiles, distributions
Case review	For selected high-risk providers, can the dashboard show a clear and reproducible explanation for the score?	Manually investigate high risk observations	Evidence/explanation card

6. Risk Score and Explainability

Score	Label	Dashboard Meaning
0-59	Typical / Low Anomaly	Pattern is broadly consistent with peers
60-79	Elevated Anomaly	Several measures differ from peers; review if capacity allows
80-89	High Anomaly	Strongly unusual patterns; prioritize for analyst review
90-100	Very High Anomaly	Extremely unusual peer-relative pattern; review first with human interpretation

Thresholds above are presentation placeholders. Final cutoffs will be determined after examining the score distribution and should be expressed primarily as percentiles or review-priority bands.

7. Proposed Power BI Dashboard

The dashboard will be designed as an analysis workflow rather than a collection of unrelated visuals. Five pages are proposed.

Page	Purpose	Key Visuals / Interactions
1. Executive Risk Overview	Summarize volume, utilization, payments, and anomaly distribution	KPI cards, anomaly risk distribution; top specialties; state map; filters
2. AI Review Queue	Rank provider-service observations for investigation	Sortable table by Risk Score; Top-N/Top-% selector, specialty, state, HCPCS(procedure code), and place of service slicers
3. Provider in depth	Explain why a selected provider is unusual	Risk gauge; peer percentiles; procedure profile; charge/payment comparisons
4. Procedure & Peer Analysis	Compare services within appropriate peer groups	Box/percentile plots or distributions; scatterplots; peer medians; facility vs non facility comparisons
5. Model & Data Transparency	Document source, methodology, assumptions, and limitations	Data lineage; feature definitions; model settings; validation tests; disclaimer

8. Technical Architecture

CMS 2024 Provider & Service data -> Python / pandas cleaning and feature engineering -> Isolation Forest + validation -> scored provider-service table -> Power BI semantic model -> interactive investigation dashboard

9. Scope Boundaries, Risks, and Ethical Guardrails

What the Project Will Claim:

- The model identifies unusual billing patterns relative to selected peers.
- The score can prioritize records for additional analyst review.
- The dashboard helps explain why a record received a high anomaly ranking.

What the Project Will NOT Claim:

- That a high score proves fraud, abuse, waste, or criminal conduct.
- That billing differences necessarily indicate poor quality of care.
- That the public dataset represents a provider's entire medical practice.
- That the model is appropriate for automated enforcement or payment denial.

CMS itself notes several limitations of the public physician/practitioner data: it covers fee-for-service Medicare rather than a provider's entire practice, does not provide quality-of-care information, and does not fully account for differences in patient sickness

10. Planned Deliverables

1. Power BI (.pbix) dashboard with five polished pages and navigation
2. Python notebook or scripts for data cleaning, feature engineering, ML scoring, and validation
3. Exported scored analytical table used by Power BI
4. Short methodology report describing lineage, model design, features, validation, and limitations
5. GitHub repository with README, screenshots, architecture, and reproducible instructions
6. Presentation/demo explaining the business problem, AI methods, dashboard, workflow, and responsible interpretation